

Haoyang He (Alan)

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EDUCATION

MSc Advanced Materials Science, University College London

09/2025 – 09/2026

Predicted Distinction · Sustainability & materials innovation focus

BSc Physics, University of Edinburgh

09/2021 – 05/2025

Second Class Hons

- Relevant modules: Microstructural Control, Nanoscale Processing & Characterization, Material Processing & Manufacturing, Condensed Matter Physics, Thermodynamics, Statistics, Quantum Physics, Electromagnetism

EXPERIENCE

Semiconductor Optoelectronic Characterisation Project, UCL

11/2025 – Present

- Investigating how bias-induced ion migration affects internal electric fields, charge extraction and hysteresis in perovskite optoelectronic devices, contributing to next-generation clean-energy materials research.
- Analyse photocurrent amplitude, polarity and relaxation behaviour to separate ionic redistribution from fast electronic transport, building defensible, evidence-based conclusions.
- Developed ionic–electronic transport models with Python-based nonlinear fitting to extract ionic relaxation times and benchmark device behaviour.
- Perform thin-film PVD and sample preparation for optical and structural characterisation workflows.

Sales Performance & Customer Segmentation Analysis, Independent Project

09/2025

- Analysed 10,000+ transaction records using SQL (CTEs, window functions, aggregations) to evaluate YoY revenue growth, product performance and seasonality, uncovering key demand trends to inform commercial decisions.
- Delivered ABC (Pareto) and RFM segmentation to identify high-value products and customer cohorts, generating actionable insight for revenue optimisation, inventory prioritisation and targeted retention.

Biofilm Rheology Project, University of Edinburgh

01/2025 – 04/2025

- Designed and led a 10-week independent project on *Bacillus subtilis* biofilm mechanics, investigating how extra-cellular matrix (ECM) components govern viscoelasticity and yield behaviour.
- Engineered and characterised four single-gene deletion mutants against wild type using amplitude- and frequency-sweep rheometry; found TasA deletion caused a > 75% drop in storage modulus and a five-fold increase in yield strain.
- Built a reproducible Python workflow (data cleaning, interpolation, Welch's t-tests, graphing) enabling statistically robust comparison across 30+ datasets and cutting manual analysis time.
- Communicated findings establishing TasA fibres as the primary cross-linker, with PGA, EPS and BslA providing secondary mechanical support.

Blood Fluid Mechanics Intern, University of Edinburgh

06/2024 – 08/2024

- Used rotational rheometry to characterise secondary flow patterns forming in Taylor–Couette cylinders above a critical Reynolds number, systematically varying Newtonian fluid viscosity.
- Extended Ellenberger (1985) to non-Newtonian fluids, demonstrating a consistent critical modified Reynolds number across both fluid types.

Solid State Microstructures Lab Volunteer, Nanjing University

08/2023 – 09/2023

- Applied PVD and spin-coating to fabricate Fe_3O_4 thin-film substrates; operated photolithography for circuit patterning.
- Self-studied Scanning Tunnelling Microscopy (STM) and vacuum-gauge principles, including STM's role in 2D crystal growth control and atomic-scale data storage.

HYPED Shell Engineer (Hyperloop Team), University of Edinburgh

09/2022 – 07/2023

- Collaborated in a cross-functional student team to design the UK's first student-built Hyperloop test pod, targeting carbon reduction and transport efficiency; owned aerodynamic design and manufacturing of the shell.
- Selected fibreglass composite to cut shell cost by ~20%, balancing performance against budget, a commercially aware engineering trade-off.

SKILLS & INTERESTS

- **Programming:** Python (scikit-learn, pandas, NumPy), MySQL, R
- **Characterisation:** XRD, FTIR, STM, SEM, Profilometry, UV-Vis, PVD, thin-film deposition
- **Data & Analysis:** Machine Learning (regression, classification, neural networks, decision trees, random forest), A/B testing, statistical hypothesis testing; Business analysis (AARRR, funnel, cohort, seasonality, OSM)
- **Tools:** Excel, Tableau, Power BI, ImageJ, LaTeX
- **Languages:** English (fluent), Mandarin (native), German (working)
- **Memberships:** Student Member, Institute of Materials, Minerals & Mining (IOM3)
- **Leadership & Engagement:** Thrash-metal band drummer; Tour du Mont Blanc finisher; Mount Kilimanjaro summitter; 2020 China National Census data-collection volunteer